

48. Evaluate each of these expressions.

- a) $1\ 1000 \wedge (0\ 1011 \vee 1\ 1011)$
- b) $(0\ 1111 \wedge 1\ 0101) \vee 0\ 1000$
- c) $(0\ 1010 \oplus 1\ 1011) \oplus 0\ 1000$
- d) $(1\ 1011 \vee 0\ 1010) \wedge (1\ 0001 \vee 1\ 1011)$

Fuzzy logic is used in artificial intelligence. In fuzzy logic, a proposition has a truth value that is a number between 0 and 1, inclusive. A proposition with a truth value of 0 is false and one with a truth value of 1 is true. Truth values that are between 0 and 1 indicate varying degrees of truth. For instance, the truth value 0.8 can be assigned to the statement “Fred is happy,” because Fred is happy most of the time, and the truth value 0.4 can be assigned to the statement “John is happy,” because John is happy slightly less than half the time. Use these truth values to solve Exercises 49–51.

- 49. The truth value of the negation of a proposition in fuzzy logic is 1 minus the truth value of the proposition. What are the truth values of the statements “Fred is not happy” and “John is not happy”?
- 50. The truth value of the conjunction of two propositions in fuzzy logic is the minimum of the truth values of the two propositions. What are the truth values of the statements

“Fred and John are happy” and “Neither Fred nor John is happy”?

- 51. The truth value of the disjunction of two propositions in fuzzy logic is the maximum of the truth values of the two propositions. What are the truth values of the statements “Fred is happy, or John is happy” and “Fred is not happy, or John is not happy”?
- *52. Is the assertion “This statement is false” a proposition?
- *53. The n th statement in a list of 100 statements is “Exactly n of the statements in this list are false.”
 - a) What conclusions can you draw from these statements?
 - b) Answer part (a) if the n th statement is “At least n of the statements in this list are false.”
 - c) Answer part (b) assuming that the list contains 99 statements.
- 54. An ancient Sicilian legend says that the barber in a remote town who can be reached only by traveling a dangerous mountain road shaves those people, and only those people, who do not shave themselves. Can there be such a barber?

1.2 Applications of Propositional Logic

1.2.1 Introduction

Logic has many important applications to mathematics, computer science, and numerous other disciplines. Statements in mathematics and the sciences and in natural language often are imprecise or ambiguous. To make such statements precise, they can be translated into the language of logic. For example, logic is used in the specification of software and hardware, because these specifications need to be precise before development begins. Furthermore, propositional logic and its rules can be used to design computer circuits, to construct computer programs, to verify the correctness of programs, and to build expert systems. Logic can be used to analyze and solve many familiar puzzles. Software systems based on the rules of logic have been developed for constructing some, but not all, types of proofs automatically. We will discuss some of these applications of propositional logic in this section and in later chapters.

1.2.2 Translating English Sentences

There are many reasons to translate English sentences into expressions involving propositional variables and logical connectives. In particular, English (and every other human language) is often ambiguous. Translating sentences into compound statements (and other types of logical expressions, which we will introduce later in this chapter) removes the ambiguity. Note that this may involve making a set of reasonable assumptions based on the intended meaning of the sentence. Moreover, once we have translated sentences from English into logical expressions, we can analyze these logical expressions to determine their truth values, we can manipulate them, and we can use rules of inference (which are discussed in Section 1.6) to reason about them.

To illustrate the process of translating an English sentence into a logical expression, consider Examples 1 and 2.

EXAMPLE 1 How can this English sentence be translated into a logical expression?**Extra
Examples** 

“You can access the Internet from campus only if you are a computer science major or you are not a freshman.”

Solution: There are many ways to translate this sentence into a logical expression. Although it is possible to represent the sentence by a single propositional variable, such as p , this would not be useful when analyzing its meaning or reasoning with it. Instead, we will use propositional variables to represent each sentence part and determine the appropriate logical connectives between them. In particular, we let a , c , and f represent “You can access the Internet from campus,” “You are a computer science major,” and “You are a freshman,” respectively. Noting that “only if” is one way a conditional statement can be expressed, this sentence can be represented as

$$a \rightarrow (c \vee \neg f).$$


EXAMPLE 2 How can this English sentence be translated into a logical expression?

“You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old.”

Solution: Let q , r , and s represent “You can ride the roller coaster,” “You are under 4 feet tall,” and “You are older than 16 years old,” respectively. Then the sentence can be translated to

$$(r \wedge \neg s) \rightarrow \neg q.$$

There are other ways to represent the original sentence as a logical expression, but the one we have used should meet our needs. 

1.2.3 System Specifications

Translating sentences in natural language (such as English) into logical expressions is an essential part of specifying both hardware and software systems. System and software engineers take requirements in natural language and produce precise and unambiguous specifications that can be used as the basis for system development. Example 3 shows how compound propositions can be used in this process.

EXAMPLE 3 Express the specification “The automated reply cannot be sent when the file system is full” using logical connectives.**Extra
Examples** 

Solution: One way to translate this is to let p denote “The automated reply can be sent” and q denote “The file system is full.” Then $\neg p$ represents “It is not the case that the automated reply can be sent,” which can also be expressed as “The automated reply cannot be sent.” Consequently, our specification can be represented by the conditional statement $q \rightarrow \neg p$. 

System specifications should be **consistent**, that is, they should not contain conflicting requirements that could be used to derive a contradiction. When specifications are not consistent, there would be no way to develop a system that satisfies all specifications.

EXAMPLE 4 Determine whether these system specifications are consistent:

“The diagnostic message is stored in the buffer or it is retransmitted.”

“The diagnostic message is not stored in the buffer.”

“If the diagnostic message is stored in the buffer, then it is retransmitted.”

Solution: To determine whether these specifications are consistent, we first express them using logical expressions. Let p denote “The diagnostic message is stored in the buffer” and let q denote “The diagnostic message is retransmitted.” The specifications can then be written as $p \vee q$, $\neg p$, and $p \rightarrow q$. An assignment of truth values that makes all three specifications true must have p false to make $\neg p$ true. Because we want $p \vee q$ to be true but p must be false, q must be true. Because $p \rightarrow q$ is true when p is false and q is true, we conclude that these specifications are consistent, because they are all true when p is false and q is true. We could come to the same conclusion by use of a truth table to examine the four possible assignments of truth values to p and q . 

EXAMPLE 5 Do the system specifications in Example 4 remain consistent if the specification “The diagnostic message is not retransmitted” is added?

Solution: By the reasoning in Example 4, the three specifications from that example are true only in the case when p is false and q is true. However, this new specification is $\neg q$, which is false when q is true. Consequently, these four specifications are inconsistent. 

1.2.4 Boolean Searches

Links 

Logical connectives are used extensively in searches of large collections of information, such as indexes of Web pages. Because these searches employ techniques from propositional logic, they are called **Boolean searches**.

In Boolean searches, the connective *AND* is used to match records that contain both of two search terms, the connective *OR* is used to match one or both of two search terms, and the connective *NOT* (sometimes written as *AND NOT*) is used to exclude a particular search term. Careful planning of how logical connectives are used is often required when Boolean searches are used to locate information of potential interest. Example 6 illustrates how Boolean searches are carried out.

EXAMPLE 6 Web Page Searching Most Web search engines support Boolean searching techniques, which is useful for finding Web pages about particular subjects. For instance, using Boolean searching to find Web pages about universities in New Mexico, we can look for pages matching *NEW AND MEXICO AND UNIVERSITIES*. The results of this search will include those pages that contain the three words *NEW*, *MEXICO*, and *UNIVERSITIES*. This will include all of the pages of interest, together with others such as a page about new universities in Mexico. (Note that Google, and many other search engines, do require the use of “AND” because such search engines use all search terms by default.) Most search engines support the use of quotation marks to search for specific phrases. So, it may be more effective to search for pages matching “*NEW MEXICO*” *AND UNIVERSITIES*. 

Extra Examples

Next, to find pages that deal with universities in New Mexico or Arizona, we can search for pages matching *(NEW AND MEXICO OR ARIZONA) AND UNIVERSITIES*. (Note: Here the *AND* operator takes precedence over the *OR* operator. Also, in Google, the terms used for this search would be *NEW MEXICO OR ARIZONA*.) The results of this search will include all pages that contain the word *UNIVERSITIES* and either both the words *NEW* and *MEXICO* or the word *ARIZONA*. Again, pages besides those of interest will be listed. Finally, to find Web pages that deal with universities in Mexico (and not New Mexico), we might first look for pages matching *MEXICO AND UNIVERSITIES*, but because the results of this search will include pages about universities in New Mexico, as well as universities in Mexico, it might be better to search for pages matching *(MEXICO AND UNIVERSITIES) NOT NEW*. The results of this search include pages that contain both the words *MEXICO* and *UNIVERSITIES* but do not contain the word *NEW*. (In Google, and many other search engines, the word “NOT” is

replaced by the symbol “-”. In Google, the terms used for this last search would be MEXICO UNIVERSITIES -NEW.)

1.2.5 Logic Puzzles

Links

Puzzles that can be solved using logical reasoning are known as **logic puzzles**. Solving logic puzzles is an excellent way to practice working with the rules of logic. Also, computer programs designed to carry out logical reasoning often use well-known logic puzzles to illustrate their capabilities. Many people enjoy solving logic puzzles, published in periodicals, books, and on the Web, as a recreational activity.

The next three examples present logic puzzles, in increasing level of difficulty. Many others can be found in the exercises. In Section 1.3 we will discuss the n -queens puzzle and the game of Sudoku.

EXAMPLE 7

As a reward for saving his daughter from pirates, the King has given you the opportunity to win a treasure hidden inside one of three trunks. The two trunks that do not hold the treasure are empty. To win, you must select the correct trunk. Trunks 1 and 2 are each inscribed with the message “This trunk is empty,” and Trunk 3 is inscribed with the message “The treasure is in Trunk 2.” The Queen, who never lies, tells you that only one of these inscriptions is true, while the other two are wrong. Which trunk should you select to win?

Solution: Let p_i be the proposition that the treasure is in Trunk i , for $i = 1, 2, 3$. To translate into propositional logic the Queen’s statement that exactly one of the inscriptions is true, we observe that the inscriptions on Trunk 1, Trunk 2, and Trunk 3, are $\neg p_1$, $\neg p_2$, and p_2 , respectively. So, her statement can be translated to

$$(\neg p_1 \wedge \neg(\neg p_2) \wedge \neg p_2) \vee (\neg(\neg p_1) \wedge \neg p_2 \wedge \neg p_2) \vee (\neg(\neg p_1) \wedge \neg(\neg p_2) \wedge p_2).$$

Using the rules for propositional logic, we see that this is equivalent to $(p_1 \wedge \neg p_2) \vee (p_1 \wedge p_2)$. By the distributive law, $(p_1 \wedge \neg p_2) \vee (p_1 \wedge p_2)$ is equivalent to $p_1 \wedge (\neg p_2 \vee p_2)$. But because $\neg p_2 \vee p_2$ must be true, this is then equivalent to $p_1 \wedge \mathbf{T}$, which is in turn equivalent to p_1 . So the treasure is in Trunk 1 (that is, p_1 is true), and p_2 and p_3 are false; and the inscription on Trunk 2 is the only true one. (Here, we have used the concept of propositional equivalence, which is discussed in Section 1.3.)

Next, we introduce a puzzle originally posed by Raymond Smullyan, a master of logic puzzles, who has published more than a dozen books containing challenging puzzles that involve logical reasoning.

EXAMPLE 8

In [Sm78] Smullyan posed many puzzles about an island that has two kinds of inhabitants, knights, who always tell the truth, and their opposites, knaves, who always lie. You encounter two people A and B . What are A and B if A says “ B is a knight” and B says “The two of us are opposite types”?

Extra Examples

Solution: Let p and q be the statements that A is a knight and B is a knight, respectively, so that $\neg p$ and $\neg q$ are the statements that A is a knave and B is a knave, respectively.

We first consider the possibility that A is a knight; this is the statement that p is true. If A is a knight, then he is telling the truth when he says that B is a knight, so that q is true, and A and B are the same type. However, if B is a knight, then B ’s statement that A and B are of opposite

types, the statement $(p \wedge \neg q) \vee (\neg p \wedge q)$, would have to be true, which it is not, because A and B are both knights. Consequently, we can conclude that A is not a knight, that is, that p is false.

If A is a knave, then because everything a knave says is false, A 's statement that B is a knight, that is, that q is true, is a lie. This means that q is false and B is also a knave. Furthermore, if B is a knave, then B 's statement that A and B are opposite types is a lie, which is consistent with both A and B being knaves. We can conclude that both A and B are knaves. ▶

We pose more of Smullyan's puzzles about knights and knaves in Exercises 23–27. In Exercises 28–35 we introduce related puzzles where we have three types of people, knights and knaves as in this puzzle together with spies who can lie.

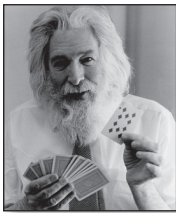
Next, we pose a puzzle known as the **muddy children puzzle** for the case of two children.

EXAMPLE 9 A father tells his two children, a boy and a girl, to play in their backyard without getting dirty. However, while playing, both children get mud on their foreheads. When the children stop playing, the father says “At least one of you has a muddy forehead,” and then asks the children to answer “Yes” or “No” to the question: “Do you know whether you have a muddy forehead?” The father asks this question twice. What will the children answer each time this question is asked, assuming that a child can see whether his or her sibling has a muddy forehead, but cannot see his or her own forehead? Assume that both children are honest and that the children answer each question simultaneously.

Solution: Let s be the statement that the son has a muddy forehead and let d be the statement that the daughter has a muddy forehead. When the father says that at least one of the two children has a muddy forehead, he is stating that the disjunction $s \vee d$ is true. Both children will answer “No” the first time the question is asked because each sees mud on the other child's forehead. That is, the son knows that d is true, but does not know whether s is true, and the daughter knows that s is true, but does not know whether d is true.

After the son has answered “No” to the first question, the daughter can determine that d must be true. This follows because when the first question is asked, the son knows that $s \vee d$ is true, but cannot determine whether s is true. Using this information, the daughter can conclude that d must be true, for if d were false, the son could have reasoned that because $s \vee d$ is true,

Links



Courtesy of Indiana University Archives

RAYMOND SMULLYAN (1919–2017) Raymond Smullyan, the son of a businessman and a homemaker, was born in Far Rockaway, Queens, New York. He dropped out of high school because he wanted to study what he was really interested in and not standard high school material. After attending Pacific College and Reed College in Oregon, he earned an undergraduate degree in mathematics at the University of Chicago in 1955. He paid his college expenses by performing magic tricks at parties and clubs, using the stage name Five-Ace Merrill. He obtained a Ph.D. in logic in 1959 at Princeton, studying under Alonzo Church. After graduating from Princeton, he taught mathematics and logic at Dartmouth College, Princeton University, Yeshiva University, and the City University of New York. He joined the philosophy department at Indiana University in 1981, where he became an emeritus professor.

Smullyan wrote many books on recreational logic and mathematics, including *Satan, Cantor, and Infinity*; *What Is the Name of This Book?*; *The Lady or the Tiger?*; *Alice in Puzzleland*; *To Mock a Mockingbird*; *Forever Undecided*; and *The Riddle of Scheherazade: Amazing Logic Puzzles, Ancient and Modern*. Because his logic puzzles are challenging, entertaining, and thought-provoking, he was considered to be a modern-day Lewis Carroll. Smullyan also wrote several books about the application of deductive logic to chess, three collections of philosophical essays and aphorisms, and several advanced books on mathematical logic and set theory. He was particularly interested in self-reference and worked on extending some of Gödel's results that show that it is impossible to write a computer program that can solve all mathematical problems. He was also particularly interested in explaining ideas from mathematical logic to the public.

Smullyan was a talented musician and often played piano with his second wife, who was a concert-level pianist. Making telescopes was one of his hobbies and he was interested in optics and stereo photography. He said, “I’ve never had a conflict between teaching and research as some people do because when I’m teaching, I’m doing research.” Smullyan is the subject of a documentary short film entitled *This Film Needs No Title*.

then s must be true, and he would have answered “Yes” to the first question. The son can reason in a similar way to determine that s must be true. It follows that both children answer “Yes” the second time the question is asked. ◀

1.2.6 Logic Circuits

Propositional logic can be applied to the design of computer hardware. This was first observed in 1938 by Claude Shannon in his MIT master’s thesis. In Chapter 12 we will study this topic in depth. (See that chapter for a biography of Shannon.) We give a brief introduction to this application here.

A **logic circuit** (or **digital circuit**) receives input signals $p_1, p_2, \dots, p_n$, each a bit [either 0 (off) or 1 (on)], and produces output signals $s_1, s_2, \dots, s_n$, each a bit. In this section we will restrict our attention to logic circuits with a single output signal; in general, digital circuits may have multiple outputs.

Complicated digital circuits can be constructed from three basic circuits, called **gates**, shown in Figure 1. The **inverter**, or **NOT gate**, takes an input bit p , and produces as output $\neg p$. The **OR gate** takes two input signals p and q , each a bit, and produces as output the signal $p \vee q$. Finally, the **AND gate** takes two input signals p and q , each a bit, and produces as output the signal $p \wedge q$. We use combinations of these three basic gates to build more complicated circuits, such as that shown in Figure 2.

In Chapter 12 we will design some useful circuits.

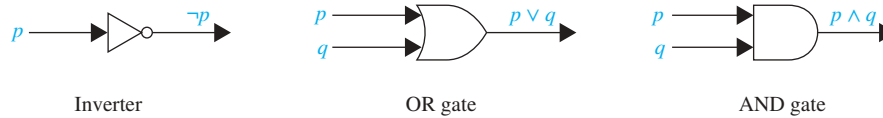


FIGURE 1 Basic logic gates.

Given a circuit built from the basic logic gates and the inputs to the circuit, we determine the output by tracing through the circuit, as Example 10 shows.

EXAMPLE 10 Determine the output for the combinatorial circuit in Figure 2.

Solution: In Figure 2 we display the output of each logic gate in the circuit. We see that the AND gate takes input of p and $\neg q$, the output of the inverter with input q , and produces $p \wedge \neg q$. Next, we note that the OR gate takes input $p \wedge \neg q$ and $\neg r$, the output of the inverter with input r , and produces the final output $(p \wedge \neg q) \vee \neg r$. ◀

Suppose that we have a formula for the output of a digital circuit in terms of negations, disjunctions, and conjunctions. Then, we can systematically build a digital circuit with the desired output, as illustrated in Example 11.

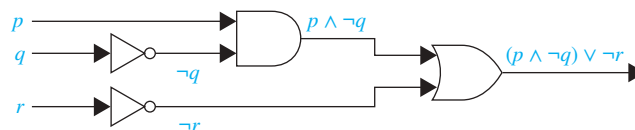


FIGURE 2 A combinatorial circuit.

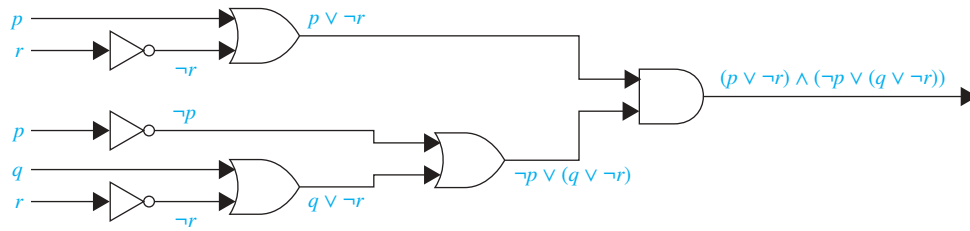


FIGURE 3 The circuit for $(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$.

EXAMPLE 11 Build a digital circuit that produces the output $(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$ when given input bits p , q , and r .

Solution: To construct the desired circuit, we build separate circuits for $p \vee \neg r$ and for $\neg p \vee (q \vee \neg r)$ and combine them using an AND gate. To construct a circuit for $p \vee \neg r$, we use an inverter to produce $\neg r$ from the input r . Then, we use an OR gate to combine p and $\neg r$. To build a circuit for $\neg p \vee (q \vee \neg r)$, we first use an inverter to obtain $\neg r$. Then we use an OR gate with inputs q and $\neg r$ to obtain $q \vee \neg r$. Finally, we use another inverter and an OR gate to get $\neg p \vee (q \vee \neg r)$ from the inputs p and $q \vee \neg r$.

To complete the construction, we employ a final AND gate, with inputs $p \vee \neg r$ and $\neg p \vee (q \vee \neg r)$. The resulting circuit is displayed in Figure 3. ◀

We will study logic circuits in great detail in Chapter 12 in the context of Boolean algebra, and with different notation.

Exercises

In Exercises 1–6, translate the given statement into propositional logic using the propositions provided.

1. You cannot edit a protected Wikipedia entry unless you are an administrator. Express your answer in terms of e : “You can edit a protected Wikipedia entry” and a : “You are an administrator.”
2. You can see the movie only if you are over 18 years old or you have the permission of a parent. Express your answer in terms of m : “You can see the movie,” e : “You are over 18 years old,” and p : “You have the permission of a parent.”
3. You can graduate only if you have completed the requirements of your major and you do not owe money to the university and you do not have an overdue library book. Express your answer in terms of g : “You can graduate,” m : “You owe money to the university,” r : “You have completed the requirements of your major,” and b : “You have an overdue library book.”
4. To use the wireless network in the airport you must pay the daily fee unless you are a subscriber to the service. Express your answer in terms of w : “You can use the wireless network in the airport,” d : “You pay the daily fee,” and s : “You are a subscriber to the service.”
5. You are eligible to be President of the U.S.A. only if you are at least 35 years old, were born in the U.S.A., or at the time of your birth both of your parents were citizens, and you have lived at least 14 years in the country. Express your answer in terms of e : “You are eligible to be President of the U.S.A.,” a : “You are at least 35 years old,” b : “You were born in the U.S.A.,” p : “At the time of your birth, both of your parents were citizens,” and r : “You have lived at least 14 years in the U.S.A.”
6. You can upgrade your operating system only if you have a 32-bit processor running at 1 GHz or faster, at least 1 GB RAM, and 16 GB free hard disk space, or a 64-bit processor running at 2 GHz or faster, at least 2 GB RAM, and at least 32 GB free hard disk space. Express your answer in terms of u : “You can upgrade your operating system,” b_{32} : “You have a 32-bit processor,” b_{64} : “You have a 64-bit processor,” g_1 : “Your processor runs at 1 GHz or faster,” g_2 : “Your processor runs at 2 GHz or faster,” r_1 : “Your processor has at least 1 GB RAM,” r_2 : “Your processor has at least 2 GB RAM,” h_{16} : “You have at least 16 GB free hard disk space,” and h_{32} : “You have at least 32 GB free hard disk space.”
7. Express these system specifications using the propositions p : “The message is scanned for viruses” and q : “The message was sent from an unknown system” together with logical connectives (including negations).
 - a) “The message is scanned for viruses whenever the message was sent from an unknown system.”

- b) "The message was sent from an unknown system but it was not scanned for viruses."
 - c) "It is necessary to scan the message for viruses whenever it was sent from an unknown system."
 - d) "When a message is not sent from an unknown system it is not scanned for viruses."
8. Express these system specifications using the propositions p : "The user enters a valid password," q : "Access is granted," and r : "The user has paid the subscription fee" and logical connectives (including negations).
- a) "The user has paid the subscription fee, but does not enter a valid password."
 - b) "Access is granted whenever the user has paid the subscription fee and enters a valid password."
 - c) "Access is denied if the user has not paid the subscription fee."
 - d) "If the user has not entered a valid password but has paid the subscription fee, then access is granted."
9. Are these system specifications consistent? "The system is in multiuser state if and only if it is operating normally. If the system is operating normally, the kernel is functioning. The kernel is not functioning or the system is in interrupt mode. If the system is not in multiuser state, then it is in interrupt mode. The system is not in interrupt mode."
10. Are these system specifications consistent? "Whenever the system software is being upgraded, users cannot access the file system. If users can access the file system, then they can save new files. If users cannot save new files, then the system software is not being upgraded."
11. Are these system specifications consistent? "The router can send packets to the edge system only if it supports the new address space. For the router to support the new address space it is necessary that the latest software release be installed. The router can send packets to the edge system if the latest software release is installed. The router does not support the new address space."
12. Are these system specifications consistent? "If the file system is not locked, then new messages will be queued. If the file system is not locked, then the system is functioning normally, and conversely. If new messages are not queued, then they will be sent to the message buffer. If the file system is not locked, then new messages will be sent to the message buffer. New messages will not be sent to the message buffer."
13. What Boolean search would you use to look for Web pages about beaches in New Jersey? What if you wanted to find Web pages about beaches on the isle of Jersey (in the English Channel)?
14. What Boolean search would you use to look for Web pages about hiking in West Virginia? What if you wanted to find Web pages about hiking in Virginia, but not in West Virginia?
15. What Google search would you use to look for Web pages relating to Ethiopian restaurants in New York or New Jersey?
16. What Google search would you use to look for men's shoes or boots not designed for work?
17. Suppose that in Example 7, the inscriptions on Trunks 1, 2, and 3 are "The treasure is in Trunk 3," "The treasure is in Trunk 1," and "This trunk is empty." For each of these statements, determine whether the Queen who never lies could state this, and if so, which trunk the treasure is in.
- a) "All the inscriptions are false."
 - b) "Exactly one of the inscriptions is true."
 - c) "Exactly two of the inscriptions are true."
 - d) "All three inscriptions are true."
18. Suppose that in Example 7 there are treasures in two of the three trunks. The inscriptions on Trunks 1, 2, and 3 are "This trunk is empty," "There is a treasure in Trunk 1," and "There is a treasure in Trunk 2." For each of these statements, determine whether the Queen who never lies could state this, and if so, which two trunks the treasures are in.
- a) "All the inscriptions are false."
 - b) "Exactly one of the inscriptions is true."
 - c) "Exactly two of the inscriptions are true."
 - d) "All three inscriptions are true."
- *19. Each inhabitant of a remote village always tells the truth or always lies. A villager will give only a "Yes" or a "No" response to a question a tourist asks. Suppose you are a tourist visiting this area and come to a fork in the road. One branch leads to the ruins you want to visit; the other branch leads deep into the jungle. A villager is standing at the fork in the road. What one question can you ask the villager to determine which branch to take?
20. An explorer is captured by a group of cannibals. There are two types of cannibals—those who always tell the truth and those who always lie. The cannibals will barbecue the explorer unless he can determine whether a particular cannibal always lies or always tells the truth. He is allowed to ask the cannibal exactly one question.
- a) Explain why the question "Are you a liar?" does not work.
 - b) Find a question that the explorer can use to determine whether the cannibal always lies or always tells the truth.
21. When three professors are seated in a restaurant, the hostess asks them: "Does everyone want coffee?" The first professor says "I do not know." The second professor then says "I do not know." Finally, the third professor says "No, not everyone wants coffee." The hostess comes back and gives coffee to the professors who want it. How did she figure out who wanted coffee?
22. When planning a party you want to know whom to invite. Among the people you would like to invite are three touchy friends. You know that if Jasmine attends, she will become unhappy if Samir is there, Samir will attend only if Kanti will be there, and Kanti will not attend unless Jasmine also does. Which combinations of these three friends can you invite so as not to make someone unhappy?

Exercises 23–27 relate to inhabitants of the island of knights and knaves created by Smullyan, where knights always tell the truth and knaves always lie. You encounter two people, *A* and *B*. Determine, if possible, what *A* and *B* are if they address you in the ways described. If you cannot determine what these two people are, can you draw any conclusions?

23. *A* says “At least one of us is a knave” and *B* says nothing.
24. *A* says “The two of us are both knights” and *B* says “*A* is a knave.”
25. *A* says “I am a knave or *B* is a knight” and *B* says nothing.
26. Both *A* and *B* say “I am a knight.”
27. *A* says “We are both knaves” and *B* says nothing.

Exercises 28–35 relate to inhabitants of an island on which there are three kinds of people: knights who always tell the truth, knaves who always lie, and spies (called normals by Smullyan [Sm78]) who can either lie or tell the truth. You encounter three people, *A*, *B*, and *C*. You know one of these people is a knight, one is a knave, and one is a spy. Each of the three people knows the type of person each of other two is. For each of these situations, if possible, determine whether there is a unique solution and determine who the knave, knight, and spy are. When there is no unique solution, list all possible solutions or state that there are no solutions.

28. *A* says “*C* is the knave,” *B* says “*A* is the knight,” and *C* says “I am the spy.”
29. *A* says “I am the knight,” *B* says “I am the knave,” and *C* says “*B* is the knight.”
30. *A* says “I am the knave,” *B* says “I am the knave,” and *C* says “I am the knave.”
31. *A* says “I am the knight,” *B* says “*A* is telling the truth,” and *C* says “I am the spy.”
32. *A* says “I am the knight,” *B* says “*A* is not the knave,” and *C* says “*B* is not the knave.”
33. *A* says “I am the knight,” *B* says “I am the knight,” and *C* says “I am the knight.”
34. *A* says “I am not the spy,” *B* says “I am not the spy,” and *C* says “*A* is the spy.”
35. *A* says “I am not the spy,” *B* says “I am not the spy,” and *C* says “I am not the spy.”

Exercises 36–42 are puzzles that can be solved by translating statements into logical expressions and reasoning from these expressions using truth tables.

36. The police have three suspects for the murder of Mr. Cooper: Mr. Smith, Mr. Jones, and Mr. Williams. Smith, Jones, and Williams each declare that they did not kill Cooper. Smith also states that Cooper was a friend of Jones and that Williams disliked him. Jones also states that he did not know Cooper and that he was out of town the day Cooper was killed. Williams also states that he saw both Smith and Jones with Cooper the day of the killing and that either Smith or Jones must have killed him. Can you determine who the murderer was if
 - a) one of the three men is guilty, the two innocent men are telling the truth, but the statements of the guilty man may or may not be true?
 - b) innocent men do not lie?

37. Steve would like to determine the relative salaries of three coworkers using two facts. First, he knows that if Fred is not the highest paid of the three, then Janice is. Second, he knows that if Janice is not the lowest paid, then Maggie is paid the most. Is it possible to determine the relative salaries of Fred, Maggie, and Janice from what Steve knows? If so, who is paid the most and who the least? Explain your reasoning.
38. Five friends have access to a chat room. Is it possible to determine who is chatting if the following information is known? Either Kevin or Heather, or both, are chatting. Either Randy or Vijay, but not both, are chatting. If Abby is chatting, so is Randy. Vijay and Kevin are either both chatting or neither is. If Heather is chatting, then so are Abby and Kevin. Explain your reasoning.
39. A detective has interviewed four witnesses to a crime. From the stories of the witnesses the detective has concluded that if the butler is telling the truth then so is the cook; the cook and the gardener cannot both be telling the truth; the gardener and the handyman are not both lying; and if the handyman is telling the truth then the cook is lying. For each of the four witnesses, can the detective determine whether that person is telling the truth or lying? Explain your reasoning.
40. Four friends have been identified as suspects for an unauthorized access into a computer system. They have made statements to the investigating authorities. Alice said, “Carlos did it.” John said, “I did not do it.” Carlos said, “Diana did it.” Diana said, “Carlos lied when he said that I did it.”
 - a) If the authorities also know that exactly one of the four suspects is telling the truth, who did it? Explain your reasoning.
 - b) If the authorities also know that exactly one is lying, who did it? Explain your reasoning.
41. Suppose there are signs on the doors to two rooms. The sign on the first door reads “In this room there is a lady, and in the other one there is a tiger”; and the sign on the second door reads “In one of these rooms, there is a lady, and in one of them there is a tiger.” Suppose that you know that one of these signs is true and the other is false. Behind which door is the lady?
- *42. Solve this famous logic puzzle, attributed to Albert Einstein, and known as the **zebra puzzle**. Five men with different nationalities and with different jobs live in consecutive houses on a street. These houses are painted different colors. The men have different pets and have different favorite drinks. Determine who owns a zebra and whose favorite drink is mineral water (which is one of the favorite drinks) given these clues: The Englishman lives in the red house. The Spaniard owns a dog. The Japanese man is a painter. The Italian drinks tea. The Norwegian lives in the first house on the left. The green house is immediately to the right of the white one. The photographer breeds snails. The diplomat lives in the yellow house. Milk is drunk in the middle house. The owner of the green house drinks coffee. The Norwegian’s house is next to the

